cMSCI: Calibrated Multimodal Semantic Coherence Index

Enhancement Summary — Three Versions Overview

1. Version Overview

Version 1: cMSCI (CLIP + CLAP)

Uses domain-specific encoders: **CLIP** [1] for text-image and **CLAP** [2] for text-audio, operating in separate 512-d embedding spaces. A trained cross-space bridge (590K params) enables image-audio comparison. The calibration pipeline has five stages:

- **Gramian volume** [3]: Measures geometric spread of normalized embedding vectors via $\det(G)^{1/2}$; low volume = high alignment.
- **Z-score normalization**: Maps raw volumes to standard scores using baseline distribution statistics, then sigmoid to [0, 1].
- **Contrastive margins**: Compares matched-pair volume against hard negatives (K=5 per channel) from a negative bank; positive margin = better than distractors.
- **Ex-MCR complementarity** [4]: Projects CLAP audio into CLIP space to measure cross-modal information uniqueness (sign-flipped: high dispersion = complementary).
- **ProbVLM adaptive weighting** [5]: BayesCap-style [6] uncertainty adapters (2 x 592K params) provide per-sample confidence for channel weight mixing.

Result: Spearman $\rho = 0.579$ ($p = 0.001$), LOO-CV $\rho = 0.425$ ($p = 0.019$), seed-robust (10/10 significant). Trained on 10,255 pairs (OmniBench + AudioCaps + augmented).

Version 2: cMSCI v2 (Gemini Embedding 2)

Replaces dual encoders with **Gemini Embedding 2** [7], a natively multimodal model producing unified 3072-d embeddings for text, image, and audio. This eliminates the need for a cross-space bridge, Ex-MCR projection, and trained probabilistic adapters. The same calibration pipeline applies, proving cMSCI is **embedding-agnostic**.

Novel contribution — Matryoshka Scale Consistency:

Gemini Embedding 2 supports **Matryoshka Representation Learning** (MRL) [8], where embeddings can be truncated to sub-dimensions (768, 1536, 3072) without retraining. We exploit this property for **training-free uncertainty estimation**: the variance of coherence scores across truncation scales measures embedding stability. High variance signals unreliable similarity, enabling adaptive channel weighting with **zero additional parameters** (vs. 1.2M parameters in v1's ProbVLM adapters).

Result: Spearman $\rho = 0.558$ ($p = 0.001$). Bootstrap test confirms v1 and v2 are statistically indistinguishable (95% CI for difference: [-0.28, +0.27]), validating that the cMSCI methodology generalizes across fundamentally different embedding architectures.

Version 3: Multi-Backbone Ensemble (v1 + v2)

Combines v1 and v2 scores via weighted average: **score = 0.4 × v1 + 0.6 × v2**. The optimal weight was determined by leave-one-out cross-validation (30/30 folds unanimously selected $w_{v1} = 0.4$). The ensemble works because domain-specific (CLIP+CLAP) and general-purpose (Gemini) encoders have **uncorrelated**

error profiles, so combining them breaks the single-model performance ceiling.

Result: Spearman ρ = **0.693** (p = 0.00002), LOO-CV ρ = 0.693. This is a **+20% improvement** over the best single backbone.

2. Results Summary

Method	Spearman ρ	p-value	Significant?	Notes
cMSCI Ensemble (v1+v2)	0.693	0.00002	Yes	Best overall
cMSCI v1 (CLIP+CLAP)	0.579	0.001	Yes	Domain-specific
cMSCI v2 (Gemini)	0.558	0.001	Yes	Unified space
LLaVA-7B (VLM Judge)	0.503	0.005	Yes	7B params at inference
CCA (joint projection)	0.409	0.025	Yes	Linear baseline
Cosine + z-norm	0.405	0.026	Yes	Simple baseline
BLIPScore + CLAPScore	0.369	0.045	Yes	Established metric
MSCI (raw cosine avg)	0.257	0.170	No	Uncalibrated
CLIPScore	0.201	0.287	No	Image-only

Table 1: Human alignment (Spearman ρ) on 30 rated samples. All three cMSCI versions significantly outperform established baselines.

3. Ablation: Contribution of Each Component

Component	Spearman ρ	p-value	Significant?
MSCI (raw cosine)	0.313	0.093	No
+ Gramian volume	0.286	0.125	No
+ z-score calibration	0.391	0.033	Yes ← critical
+ contrastive margin	0.367	0.046	Yes
+ Ex-MCR complementarity	0.399	0.029	Yes
+ adaptive weighting (full)	0.579	0.001	Yes

Table 2: Incremental ablation of cMSCI v1 components. Z-score calibration is the critical transition from non-significant to significant correlation.

4. Novel Contributions

- **Geometric calibration pipeline:** First metric integrating Gramian volume, contrastive margins, cross-modal complementarity, and adaptive uncertainty in a single coherent framework for three-way (text-image-audio) evaluation.
- **Embedding-agnostic validation:** Same methodology validated on fundamentally different architectures (dual-space CLIP+CLAP vs. unified Gemini), proving the pipeline generalizes beyond

backbone choice.

- **Matryoshka Scale Consistency:** Novel training-free uncertainty estimation via MRL truncation variance. Replaces 1.2M trained parameters with zero additional cost.
- **Multi-backbone ensemble:** Demonstrates that complementary error profiles from heterogeneous encoders yield a 20% improvement, breaking the single-model ceiling.

5. External Validation

Benchmark	Metric	Result
AudioCaps (1,000 samples)	AUC	0.969
AudioCaps (1,000 samples)	Accuracy	90.9%
Seed robustness (10 seeds)	$\rho \pm$ std	0.519 ± 0.000
Seed robustness (10 seeds)	Significant	10/10

References

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